

SHETH L.U.J AND SIR M.V. COLLEGE

SUBJECT NAME: **Data Analysis with SAS / SPSS / R**

PRACTICAL 5

AIM: Sorting data using arrange() in R.

>COMMANDS

We need the 'dplyr' package for the 'arrange' function!

library(dplyr)

library(readr)

Just making sure we're using our AI Jobs data!

job_title <-

read_csv("AI_Impact_on_Jobs_2030.csv")

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Remember, 'arrange' sorts smallest-to-largest by default.

Example 1: Find the lowest paying jobs by sorting salary.

Sort salary from lowest to highest (ascending).

```
job_title_sorted_salary <- job_title |>  
  arrange(Average_Salary)
```

Check the top 5 to see the lowest salaries.

```
head(job_title_sorted_salary, 5)
```

Example 2: Which jobs face the highest automation risk?

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```
# Sort automation probability from highest to lowest using desc().
```

```
job_title_sorted_auto_desc <- job_title |>
```

```
arrange(desc(Automation_Probability_2030))
```

```
# The top 5 should show the highest automation probabilities.
```

```
head(job_title_sorted_auto_desc, 5)
```

```
# Example 3: Combine two criteria for a detailed sort.
```

```
# 1. Primary Sort: Alphabetical Risk Category (High, Low, Medium).
```

```
# 2. Secondary Sort: Highest salary first within each risk category.
```

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```
job_title_multi_sort <- job_title |>  
  arrange(Risk_Category,  
  desc(Average_Salary))
```

Check the top 10 rows to see the mixed results.

```
head(job_title_multi_sort, 10)
```

Example 4: The power move - filter THEN sort!

Find senior employees (> 25 yrs) and sort them by the lowest Tech Growth Factor.

```
high_exp_by_tech_growth <- job_title |>  
  filter(Years_Experience > 25) |>  
  arrange(Tech_Growth_Factor)
```

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```
cat("Here are the Top 5 senior jobs with the  
lowest tech growth impact:\n")  
# Select just the relevant columns to confirm  
the sort order.  
print(high_exp_by_tech_growth |>  
select(Years_Experience,  
Tech_Growth_Factor, Average_Salary) |>  
head(5))
```

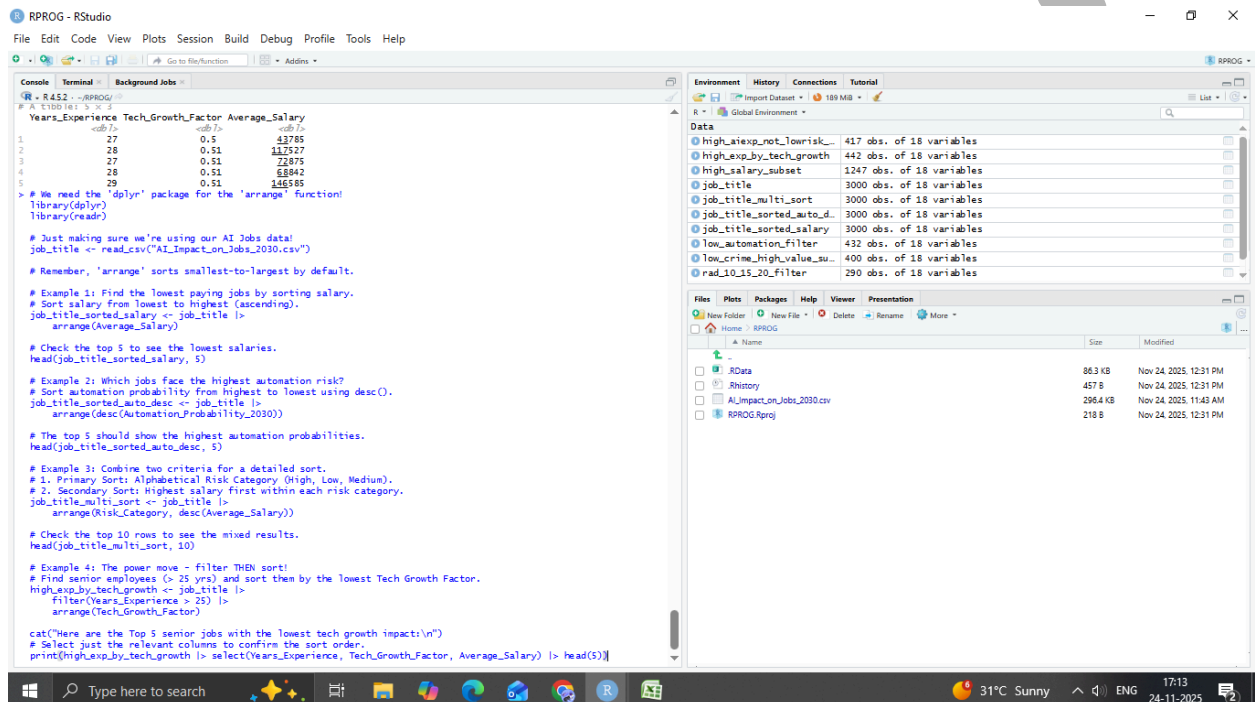
OUTPUT:

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The screenshot displays the RStudio environment. The console on the left shows the following R code and its output:

```
R > library(dplyr)
R > library(readr)
R > # We need the 'dplyr' package for the 'arrange' function!
R > library(dplyr)
R > library(readr)
R > # Just making sure we're using our AI Jobs data!
R > job_title <- read_csv("AI_Impact_on_Jobs_2030.csv")
R > # Remember, 'arrange' sorts smallest-to-largest by default.
R > # Example 1: Find the lowest paying jobs by sorting salary.
R > # Sort salary from lowest to highest (ascending).
R > job_title_sorted_salary <- job_title %>%
R >   arrange(Average_Salary)
R > # Check the top 5 to see the lowest salaries.
R > head(job_title_sorted_salary, 5)
R > # Example 2: Which jobs face the highest automation risk?
R > # Sort automation probability from highest to lowest using desc().
R > job_title_sorted_auto_desc <- job_title %>%
R >   arrange(desc(Automation_Probability_2030))
R > # The top 5 should show the highest automation probabilities.
R > head(job_title_sorted_auto_desc, 5)
R > # Example 3: Combine two criteria for a detailed sort.
R > # 1. Primary Sort: Alphabetical Risk Category (High, Low, Medium).
R > # 2. Secondary Sort: Highest salary First within each risk category.
R > job_title_multi_sort <- job_title %>%
R >   arrange(Risk_Category, desc(Average_Salary))
R > # Check the top 10 rows to see the mixed results.
R > head(job_title_multi_sort, 10)
R > # Example 4: The power move - Filter THEN sort!
R > # Find senior employees (> 25 yrs) and sort then by the lowest Tech Growth Factor.
R > high_exp_by_tech_growth <- job_title %>%
R >   filter(Years_Experience > 25) %>%
R >   arrange(Tech_Growth_Factor)
R > cat("Here are the Top 5 senior jobs with the lowest tech growth impact:\n")
R > # Select just the relevant columns to confirm the sort order.
R > print(high_exp_by_tech_growth %>% select(Years_Experience, Tech_Growth_Factor, Average_Salary) %>% head(5))
```

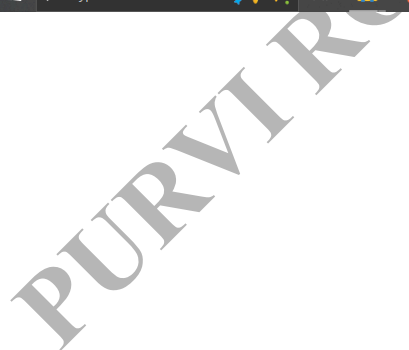
The Environment pane on the right lists the following datasets:

Dataset Name	Observations	Variables
High_Alexp_not_lowrisk_	417	18
High_exp_by_tech_growth	442	18
High_salary_subset	1247	18
job_title	3000	18
job_title_multi_sort	3000	18
job_title_sorted_auto_d.	3000	18
job_title_sorted_salary	3000	18
low_automation_filter	432	18
Low_crime_high_value_su...	400	18
rad_10_15_20_filter	290	18

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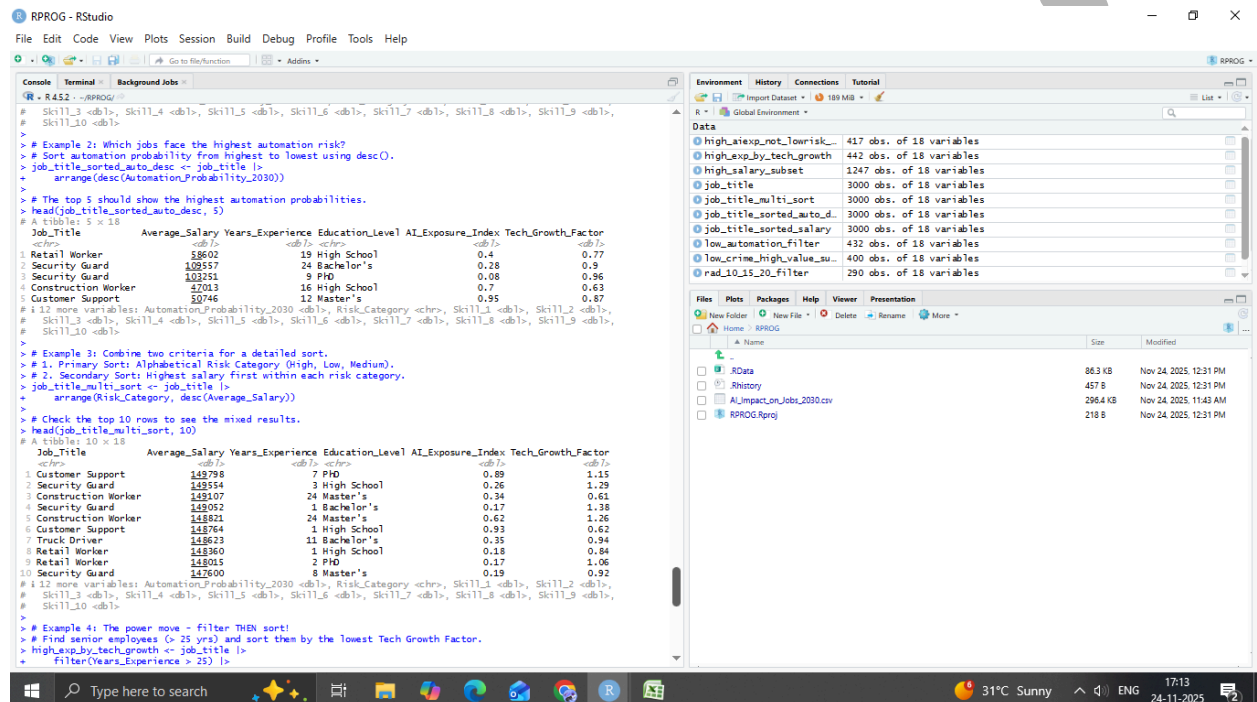
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The screenshot displays the RStudio environment with the following components:

- Console:** Contains R code for data manipulation and a printed table of results.
- Environment:** Lists data objects such as `High_Aiexp_not_lowrisk`, `High_exp_by_tech_growth`, and `High_salary_subset`.
- Files:** Shows a file explorer with items like `RData`, `Rhistory`, and `AI_Impact_on_Jobs_2030.csv`.

Console Output:

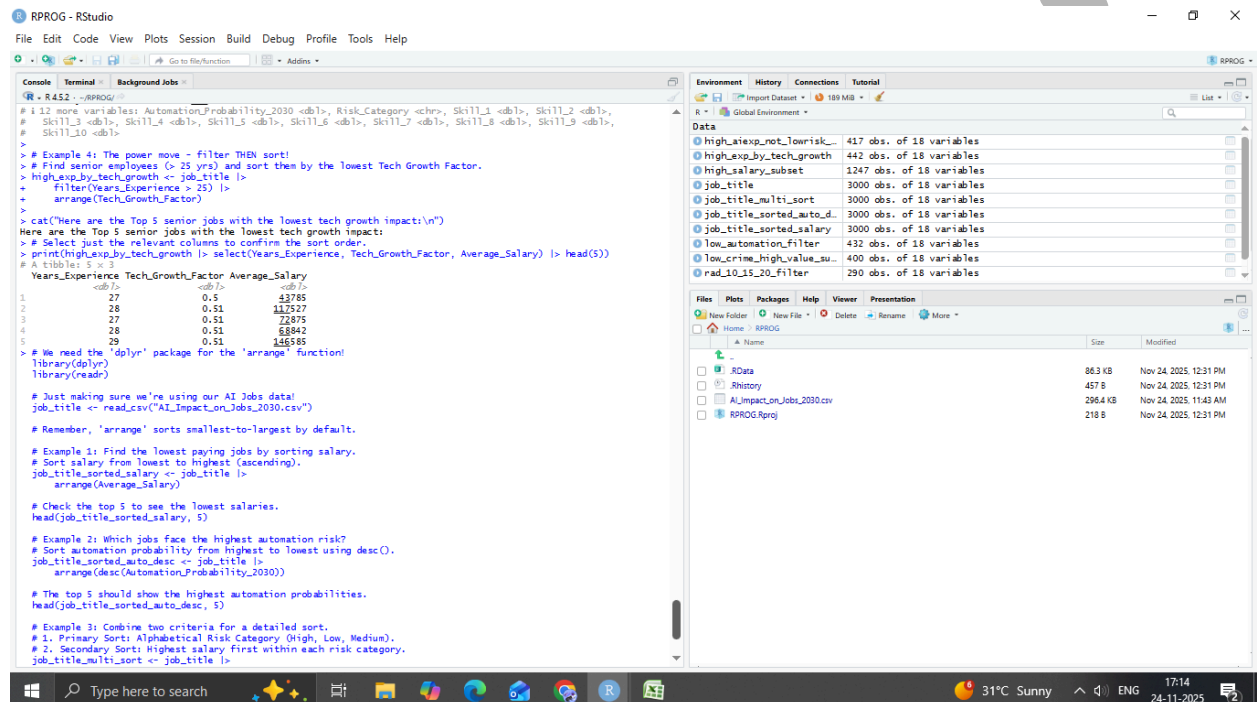
```
# Skill1_3 <dbl>, Skill1_4 <dbl>, Skill1_5 <dbl>, Skill1_6 <dbl>, Skill1_7 <dbl>, Skill1_8 <dbl>, Skill1_9 <dbl>,
# Skill1_10 <dbl>
>
> # Example 2: Which jobs face the highest automation risk?
> # Sort automation probability from highest to lowest using desc().
job_title_sorted_auto_desc <- job_title |>
+ arrange(desc(Automation_Probability_2030))
>
> # The top 5 should show the highest automation probabilities.
> head(job_title_sorted_auto_desc, 5)
# A tibble: 5 x 18
  Job_Title Average_Salary Years_Experience Education_Level AI_Exposure_Index Tech_Growth_Factor
<chr> <dbl> <dbl> <chr> <dbl> <dbl>
1 Retail Worker 18602 19 High School 0.4 0.77
2 Security Guard 102557 24 Bachelor's 0.28 0.9
3 Security Guard 103251 9 PhD 0.08 0.96
4 Construction Worker 12013 16 High School 0.7 0.63
5 Customer Support 10746 12 Master's 0.95 0.87
# 12 more variables: Automation_Probability_2030 <dbl>, Risk_Category <chr>, Skill1_1 <dbl>, Skill1_2 <dbl>,
# Skill1_3 <dbl>, Skill1_4 <dbl>, Skill1_5 <dbl>, Skill1_6 <dbl>, Skill1_7 <dbl>, Skill1_8 <dbl>, Skill1_9 <dbl>,
# Skill1_10 <dbl>
>
> # Example 3: Combine two criteria for a detailed sort.
> # 1. Primary Sort: Alphabetical Risk Category (High, Low, Medium).
> # 2. Secondary Sort: Highest salary first within each risk category.
job_title_multi_sort <- job_title |>
+ arrange(Risk_Category, desc(Average_Salary))
>
> # Check the top 10 rows to see the mixed results.
> head(job_title_multi_sort, 10)
# A tibble: 10 x 18
  Job_Title Average_Salary Years_Experience Education_Level AI_Exposure_Index Tech_Growth_Factor
<chr> <dbl> <dbl> <chr> <dbl> <dbl>
1 Customer Support 149798 7 PhD 0.89 1.15
2 Security Guard 149554 3 High School 0.26 1.29
3 Construction Worker 149107 24 Master's 0.34 0.61
4 Security Guard 149052 1 Bachelor's 0.17 1.38
5 Construction Worker 148821 24 Master's 0.62 1.26
6 Customer Support 148764 1 High School 0.93 0.62
7 Truck Driver 148623 11 Bachelor's 0.35 0.94
8 Retail Worker 148360 1 High School 0.18 0.84
9 Security Guard 148015 2 PhD 0.17 1.06
10 Security Guard 147600 8 Master's 0.19 0.92
# 12 more variables: Automation_Probability_2030 <dbl>, Risk_Category <chr>, Skill1_1 <dbl>, Skill1_2 <dbl>,
# Skill1_3 <dbl>, Skill1_4 <dbl>, Skill1_5 <dbl>, Skill1_6 <dbl>, Skill1_7 <dbl>, Skill1_8 <dbl>, Skill1_9 <dbl>,
# Skill1_10 <dbl>
>
> # Example 4: The power move - filter THEN sort!
> # Find senior employees (> 25 yrs) and sort then by the lowest Tech Growth Factor.
HighExp_by_tech_growth <- job_title |>
+ filter(Years_Experience > 25) |>
```

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The screenshot displays the RStudio environment with the following components:

- Console:** Contains R code for data manipulation and a printed table of the top 5 senior jobs with the lowest tech growth impact.
- Environment:** Lists data objects created during the session.
- Files:** Shows the file explorer with the project directory.

```
R 4.52.0 - RPROG/
File Edit Code View Plots Session Build Debug Profile Tools Help
Go to file/function Address
Console Terminal Background Jobs
# I 12 more variables: AutomationProbability_2030 <dbl>, Risk_Category <chr>, Skill_1 <dbl>, Skill_2 <dbl>,
# Skill_3 <dbl>, Skill_4 <dbl>, Skill_5 <dbl>, Skill_6 <dbl>, Skill_7 <dbl>, Skill_8 <dbl>, Skill_9 <dbl>,
# Skill_10 <dbl>
>
> # Example 4: The power move - filter THEN sort!
> # Find senior employees (> 25 yrs) and sort then by the lowest Tech Growth Factor.
> high_exp_by_tech_growth <- job_title |>
+ filter(Years_Experience > 25) |>
+ arrange(Tech_Growth_Factor)
>
> cat('Here are the Top 5 senior jobs with the lowest tech growth impact:\n')
Here are the Top 5 senior jobs with the lowest tech growth impact:
> # Select just the relevant columns to confirm the sort order.
> print(high_exp_by_tech_growth |> select(Years_Experience, Tech_Growth_Factor, Average_Salary) |> head(5))
# A tibble: 5 x 3
  Years_Experience Tech_Growth_Factor Average_Salary
      <dbl>         <dbl>         <dbl>
1          27          0.5         43785
2          28          0.51        112527
3          27          0.51        22875
4          26          0.51        68942
5          29          0.51        146585
>
> # We need the 'dplyr' package for the 'arrange' function!
library(dplyr)
library(readr)

# Just making sure we're using our AI Jobs data!
job_title <- read_csv("AI_Impact_on_Jobs_2030.csv")

# Remember, 'arrange' sorts smallest-to-largest by default.

# Example 1: Find the lowest paying jobs by sorting salary.
# Sort salary from lowest to highest (ascending).
job_title_sorted_salary <- job_title |>
+ arrange(Average_Salary)

# Check the top 5 to see the lowest salaries.
head(job_title_sorted_salary, 5)

# Example 2: Which jobs face the highest automation risk?
# Sort automation probability from highest to lowest using desc().
job_title_sorted_auto_desc <- job_title |>
+ arrange(desc(AutomationProbability_2030))

# The top 5 should show the highest automation probabilities.
head(job_title_sorted_auto_desc, 5)

# Example 3: Combine two criteria for a detailed sort.
# 1. Primary Sort: Alphabetical Risk Category (High, Low, Medium).
# 2. Secondary Sort: Highest salary first within each risk category.
job_title_multi_sort <- job_title |>
```

Environment:

Object	Size	Variables
high_exp_by_tech_growth	417 obs.	18 variables
high_exp_by_tech_growth	442 obs.	18 variables
high_salary_subset	1247 obs.	18 variables
job_title	3000 obs.	18 variables
job_title_multi_sort	3000 obs.	18 variables
job_title_sorted_auto_desc	3000 obs.	18 variables
job_title_sorted_salary	3000 obs.	18 variables
low_automation_filter	432 obs.	18 variables
low_crime_high_value_subset	400 obs.	18 variables
rad_10_15_20_filter	290 obs.	18 variables

Files:

Name	Size	Modified
AI_Impact_on_Jobs_2030.csv	296.4 KB	Nov 24, 2025, 11:43 AM
RPROG.Rproj	218 B	Nov 24, 2025, 12:31 PM

Windows taskbar: 31°C Sunny, 17:14, 24-11-2025

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